peaches: ChEES-adapted ensemble HMC

A reproducible benchmark of the Hamiltonian walk move vs. `emcee` and NUTS

Abstract

PEACHES is the *Hamiltonian walk move* — an affine-invariant ensemble HMC proposal whose metric is the empirical covariance of the live ensemble — with its integration length tuned by the **ChEES** criterion (Change in the Estimator of the Expected Square; Hoffman, Radul & Sountsov, AISTATS 2021) and its step size by Nesterov dual averaging (Hoffman & Gelman, JMLR 2014). We benchmark it against `emcee` (the gradient-free affine-invariant stretch move; Goodman & Weare 2010, Foreman-Mackey et al. 2013) and NUTS (`numpyro`) on controlled targets with exact moments, over three axes: dimensional scaling, conditioning vs. curvature, and warmup robustness. Every run is checked for convergence (rank-normalized split $\hat{R}$), and every reported efficiency is a median over the *converged* seeds only; all configurations and commands are given for exact reproduction.

Overview

PEACHES combines three ingredients, and the three benchmarks below each isolate one. **(i)** Its preconditioner is the *empirical covariance of the live ensemble*, recomputed every step, so — exactly like `emcee` — it is affine-invariant: linear ill-conditioning is handled *for free*, with no mass matrix to fit (Exp. 2a) and no mass-matrix warmup to pay for (Exp. 3). **(ii)** It integrates *gradient* Hamiltonian dynamics, so — unlike `emcee` — it crosses position-dependent curvature (Exp. 2b) and follows the optimal-HMC dimensional-scaling law (Exp. 1). **(iii)** Its trajectory length is tuned online by ChEES and its step size by dual averaging, so it is turn-key. The four axes below — dimension, conditioning, curvature, warmup — are exactly the four places single-strategy samplers fail, and PEACHES is the only one here that clears all four at once:

Axis (experiment)	what it stresses	PEACHES	`emcee`	NUTS
Dimension (Exp. 1)	cost growth with d	$d^{-1/4}$ (optimal)	d^{-1}	$d^{-1/4}$
Conditioning (Exp. 2a)	linear anisotropy κ	flat (free)	flat (free)	flat *only* w/ mass adapt.
Curvature (Exp. 2b)	nonlinear ridge	crosses it	fails ($\hat{R}\to 2$)	crosses it
Warmup (Exp. 3)	tuning cost	flat from w=50	—	needs $w \gtrsim 500$

`emcee` is affine-invariant but gradient-free, so it dies on both dimension and curvature; NUTS integrates gradients but must *fit* a mass matrix, so it needs a long warmup and, without it, loses conditioning-robustness. PEACHES inherits the ensemble metric from the first and gradient dynamics from the second, and pays for neither.

1 Methods

peaches: a ChEES-adapted Hamiltonian walk move. Let the ensemble be W walkers $q_{1:W} \in \mathbb{R}^d$ with target $\pi(q) \propto e^{-U(q)}$. The *Hamiltonian walk move* ("h-walk") proposes a Hamilto-

nian trajectory for a subset of walkers using a momentum that lives in the span of the *complement* ensemble: with C the centered complement walkers (rows $q_j - \bar{q}$), momentum p is a $W \times W$ matrix and the leapfrog updates are

$$q \leftarrow q + \epsilon\, p\, C, \qquad p \leftarrow p - \epsilon\, \nabla U(q)\, C^\top,$$

so positions move only within the affine subspace spanned by the ensemble and the effective metric is the empirical ensemble covariance $\Sigma = \mathrm{cov}(C)$. This makes the move *affine-invariant*: it is unchanged under any invertible linear reparametrization of q, so its efficiency is independent of the target's conditioning. (A complementary "h-side" move proposes along a single random ensemble difference direction; both are Metropolis-corrected with the standard HMC accept ratio.) The preconditioner is recomputed from the live ensemble *every iteration* — there is no separately-fit mass matrix.

Adaptation (warmup only). Three independently-toggleable adapters run during warmup and are then frozen:

1. *Initial step size* via a heuristic line search (`find_init_step_size`): double/halve ϵ until the first trajectory's acceptance brackets the target.

2. *Step size* ϵ via Nesterov dual averaging (Hoffman & Gelman 2014) toward a target acceptance (we pin 0.8 for the scaling law). The log-step update uses gain $\sqrt{t}/\gamma$ on the running acceptance error $\bar{H}$; the realized acceptance is the harmonic mean of per-walker $\min(1, e^{\log \alpha})$.

3. *Integration length* $T = \epsilon L$ via the **ChEES** criterion (Hoffman et al. 2021). ChEES does ADAM ascent on the expected change in squared jump distance: the per-step gradient signal is $g \propto h_t\, T\, (\|c_{\mathrm{pro}}\|^2_\Sigma - \|c_{\mathrm{cur}}\|^2_\Sigma)\, (c_{\mathrm{pro}}^\top \Sigma^{-1} v)$, where c are centered positions, v the trajectory velocity, h_t a Halton sequence, and $\|\cdot\|_\Sigma$ the *affine-invariant* Σ^{-1} norm (matching the move's metric). ADAM (β_1=0, β_2=0.95, lr 0.025, per-step clip $\pm$0.35) updates $\log T$; the realized length is Halton-jittered, $L = \lceil T/\epsilon \rceil$ capped at $L_{\max}$.

Setting all three flags off reduces warmup to a plain burn-in at a fixed (ϵ, L). PEACHES thus realizes *optimally-tuned* ensemble HMC: the metric is free (the ensemble), the step size hits a target acceptance, and the trajectory length maximizes squared jump distance.

Baselines. `emcee` is the gradient-free affine-invariant stretch move (Goodman & Weare 2010; `emcee`, Foreman-Mackey et al. 2013): it is exactly affine-invariant like the h-walk but uses *no* gradient. NUTS is `numpyro`'s implementation with window mass-matrix adaptation; we use the mass matrix appropriate to each experiment (§below) and also an *ablation* with the mass fixed to the identity (`adapt_mass_matrix=False`) to isolate the role of preconditioning.

Targets, metric, convergence. Targets are the isotropic Gaussian $N(0, I_d)$, the rotated ill-conditioned Gaussian (condition number $\kappa = e^{\log \kappa}$), and the Rosenbrock banana — all with exact moments. Efficiency is *production-only* ESS per **target evaluation** (worst coordinate): this counts one *gradient* per leapfrog step for the gradient samplers (PEACHES, NUTS) and one *log-density* per walker per iteration for gradient-free `emcee` — the fair per-oracle-call cost in each case. Throughout we use the *ensemble-mean* ESS estimator: the integrated autocorrelation of the walker-averaged series, rescaled by the pooled variance ratio. This is the natural estimator for an interacting ensemble and is $O(dN \log N)$ — no per-walker FFT — so it stays cheap even at the thousands

of walkers the scaling sweep reaches (it agrees with ArviZ's rank-normalized ESS to within the mild antithetic boost of well-tuned HMC, $\lesssim 25\%$). Ensemble methods use $n_{\text{walkers}} = 8d$ — a *constant* 8 walkers per dimension (floor 64), so the empirical covariance and the ChEES trajectory-length estimate are equally well-resolved at every d (§2) — except the fixed-d warmup experiment, where any full-rank count suffices (400 at d=100). Ensemble runs also use a stretch pre-warmup (§6). **Convergence is enforced throughout**: each per-dimension point is the median over its *converged* seeds (rank-normalized split $\hat{R} \leq 1.03$; relaxed slightly from 1.01 for seed-median plus thinning noise, all genuine failures sit at $\hat{R} \geq 1.04$), a stuck seed never enters the median, and a point is drawn solid only if a *majority* of its seeds converged (hollow otherwise — a hollow point's ESS/eval is not a meaningful efficiency).

2 Experiment 1 — dimensional scaling

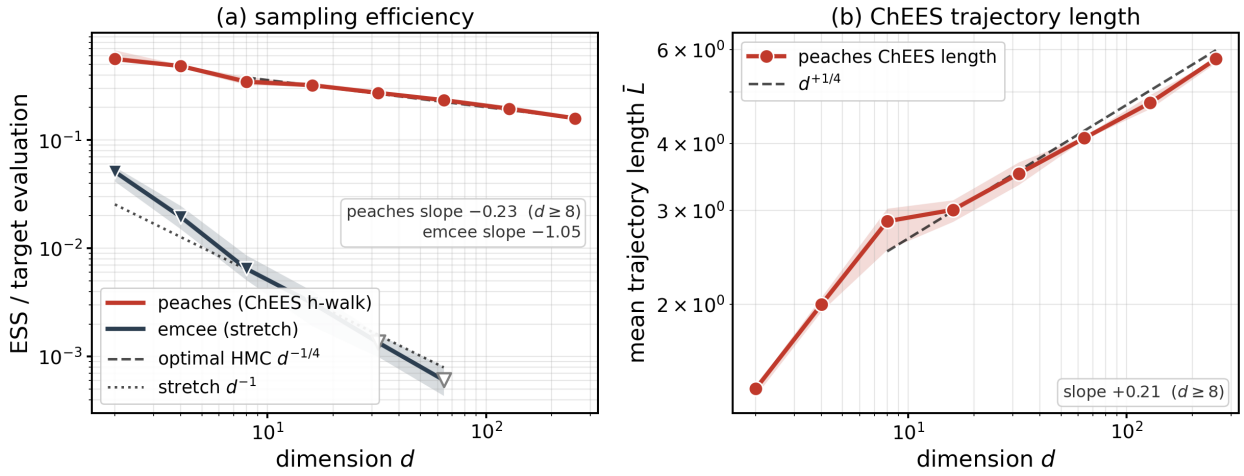


Figure 1: Isotropic Gaussian $N(0, I_d)$, d = 2–256, constant $n_{\text{walkers}} = 8d$ (floor 64), 5 seeds (3 at d=256), ensemble-mean ESS. Median over the converged seeds; band = seed min–max. **(a)** sampling efficiency: PEACHES is a clean monotone optimal-HMC $d^{-1/4}$ law across the whole range (fitted slope −0.23 for $d \geq 8$, theory −0.25), while `emcee` falls as d^{-1} (fitted −1.05) — an exponentially widening gap ($\sim$11× at d=2, $\sim$380× at d=64). **(b)** the mechanism: PEACHES's ChEES-tuned mean trajectory length grows smoothly as $d^{+1/4}$ (fitted +0.21), the signature of the $\epsilon \sim d^{-1/4}$ step-size law. Holding the walkers-per-dimension ratio fixed at 8 is what makes both panels smooth: the ChEES length estimate is then equally well-resolved at every d, with no low-d over-integration. (NUTS also achieves $d^{-1/4}$ and is omitted here so the peaches-vs-emcee contrast reads cleanly; it is benchmarked in §4–5.)

Setup. Target $N(0, I_d)$ (`--no-rotate --log-cond 0`) for $d \in \{2, 4, \ldots, 256\}$, constant $n_{\text{walkers}} = 8d$ (floor 64), 5 seeds (3 at d=256; tag `scale3`). PEACHES pins `--target-accept 0.8` so the step-size law holds at a fixed acceptance; stretch pre-warmup w_s=100, warmup 4000 (long enough for dual averaging to converge), production 5000 (3000 at $d \geq 128$; ESS/eval is a ratio, insensitive to production length once converged), $L_{\max} = 512$. Runs are scored with the fast ensemble-mean ESS (`--fast-ess`), which makes even the 2048-walker d=256 cell cheap. `emcee` mixes far more slowly, so it is run much longer — warmup 3000, production 1.2×10^4 — and shown only where it reaches

$\hat{R} \leq 1.03$ ($d \leq 64$). On the paired Rosenbrock (d=2–32, Fig. 2) PEACHES uses the same constant $8d$ walker rule (floor 64), warmup 8000, 8 seeds (tag `rz8`), and `emcee` the same walkers (tag `rz8`).

Result. Optimally-tuned HMC needs $\epsilon \sim d^{-1/4}$ at fixed acceptance, hence $L \sim d^{1/4}$ and ESS/eval $\sim d^{-1/4}$ (Beskos et al. 2013). PEACHES realizes this cleanly across the whole range: the fitted efficiency slope is -0.23 ($d \geq 8$; theory -0.25) and $\bar{L}$ grows as $d^{+1/4}$ (fitted $+0.21$, $\bar{L}$ from 1.4 at d=2 to 5.8 at d=256). The key is holding the walkers-per-dimension ratio *constant* at 8: the optimal length is small in low dimensions ($d^{1/4} < 2$), and resolving it requires that the empirical covariance — and hence the ESJD gradient ChEES ascends — be well-estimated. At a fixed $8d$ walkers it is well-estimated at *every* d, so ChEES adapts the correct short $\bar{L}$ throughout and the efficiency is a smooth monotone decline with no low-d hump. (At a fixed $4d$ the covariance is too noisy in low d; ChEES then over-integrates — $\bar{L} \approx 6$ where ≤ 3 is optimal — and the per-eval efficiency is spuriously suppressed, producing a low-d artefact.) `emcee`'s gradient-free stretch move scales as ESS/eval $\sim d^{-1}$ (fitted -1.05 over the converged $d \in [8, 64]$); already at d=64 PEACHES is $\sim$380× more efficient per target evaluation, the gap widens exponentially in d, and `emcee` cannot be converged at all beyond d=64 in a feasible run.

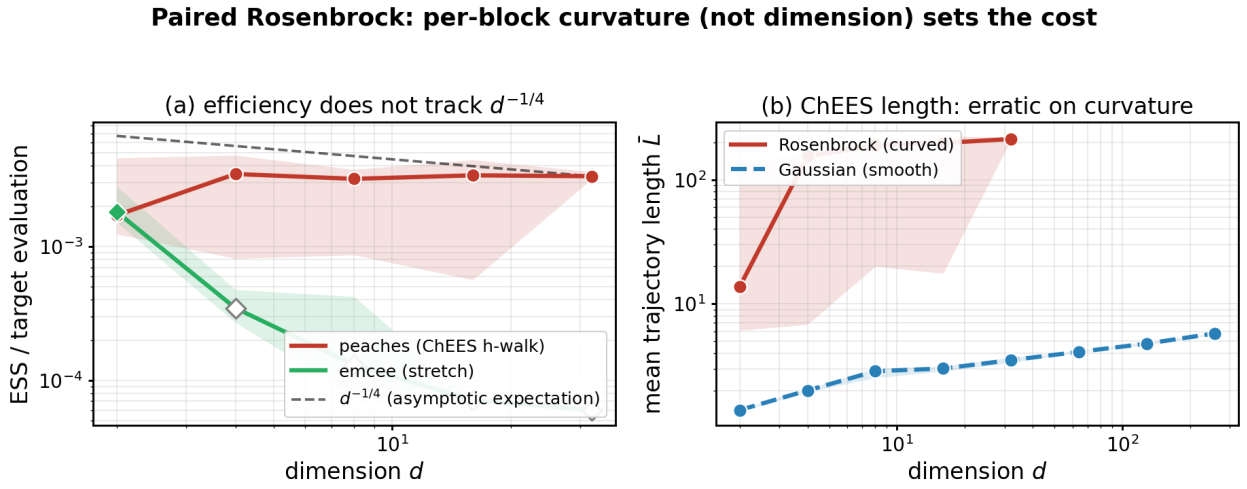


Figure 2: Dimensional scaling on the *paired* Rosenbrock (a product of $d/2$ identical banana pairs). Median over the converged seeds, band = seed min–max; hollow = a minority of seeds converged. **(a)** PEACHES stays efficient (ESS/eval $\sim 10^{-3}$) and converges, but does *not* fall as $d^{-1/4}$ (dashed reference) — it drifts upward; `emcee` fails to converge for $d \geq 4$. **(b)** the mechanism: PEACHES's ChEES length is large and erratic on the curved banana ($\bar{L} \sim$ 10–220, bimodal across seeds) but smooth on the Gaussian ($\bar{L} \sim$ 2–9, short in low d, growing as $d^{1/4}$); the curvature, not the dimension, sets the cost.

Rosenbrock — the law is masked by curvature. The paired Rosenbrock is formally a product of $d/2$ identical banana pairs, so in principle it obeys the same $d^{-1/4}$ law. It does not over the accessible range $d \leq 32$ (Fig. 2a). PEACHES converges (where `emcee` cannot) and stays efficient (ESS/eval $\sim 10^{-3}$), but the curve is seed-variable and does not fall as $d^{-1/4}$. The cause (Fig. 2b) is the per-block curvature: it gives ChEES no single good integration length, so the adapted $\bar{L}$ is *bimodal* — a long trajectory that traverses a banana ($\bar{L} \sim 200$) or a short stuck one ($\bar{L} \sim 15$) — versus the Gaussian's smooth $\bar{L}$. This bimodality is not a mere nuisance: it is a genuine,

reproducible **failure mode of trajectory-length adaptation on curved targets**, it sets the whole cost, and — as we show next — it is invisible to the usual $\hat{R}$ convergence check. We therefore dissect it in §3; the robust curvature statement is the qualitative 2×2 of §4, not a power law.

3 A ChEES failure mode: trajectory-length bistability on curvature

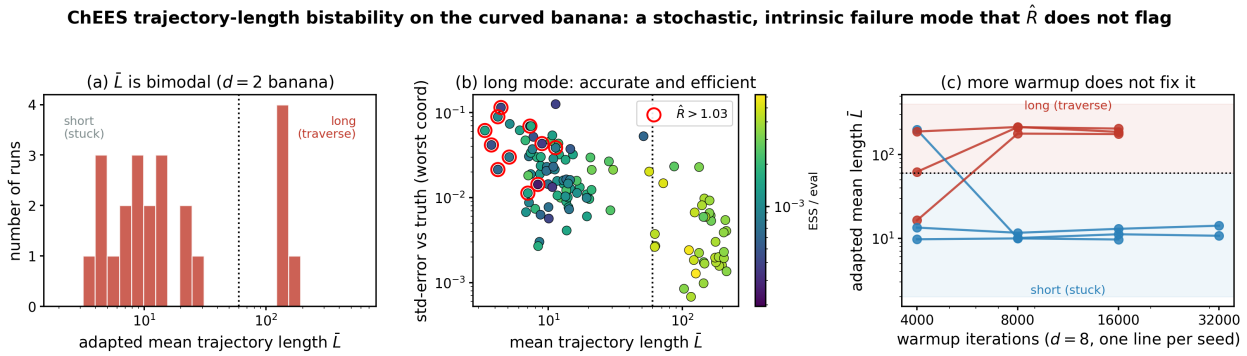


Figure 3: ChEES trajectory-length *bistability* on the banana, characterised on the cheap d=2 single banana (which already carries the curvature) and corroborated at d=8. **(a)** across many seeds the adapted $\bar{L}$ is sharply *bimodal* — a short "stuck" mode ($\bar{L} \lesssim 30$) and a long "ridge-traversing" mode ($\bar{L} \gtrsim 100$), with a clean gap. **(b)** the long mode is better on *both* axes at once — lower error vs. the exact moments *and* higher ESS/eval (colour) — so this is a good-vs-stuck split, not a speed/accuracy trade-off; crucially, almost every run passes rank-split $\hat{R} \leq 1.03$ (only the few red-ringed points fail), so $\hat{R}$ **does not separate the modes**. **(c)** tracking each d=8 seed's adapted $\bar{L}$ as warmup grows $4000 \to 32000$: seeds sit in the long (red) or short (blue) mode and *stay* there — one even drifts long→short with *more* warmup — so the split is intrinsic and stochastic, not an under-tuning artefact that a longer warmup removes.

Setup. d=2 Rosenbrock (one banana), 64 walkers, `--max-L 512`, `--target-accept 0.8`, production 8000, scored with the ensemble ESS; up to 24 seeds per cell (tags `b2_ww_is`). We also varied the initialisation scale over $\{0.5, 1, 2, 4\}$, which did not systematically shift the long/short fraction. The d=8 corroboration (Fig. 3c) fixes 256 walkers and sweeps warmup $\{4000, 8000, 16000, 32000\}$ across 6 seeds (tag `rwarm_ww`).

Result — an intrinsic, $\hat{R}$-invisible bistability. On the curved ridge the ChEES objective (expected squared jump distance under the ensemble metric) has *two* self-consistent fixed points, and the adapted length collapses onto one or the other (Fig. 3a, a clean bimodal histogram). The two are *not* equivalent: the long mode traverses a banana in one trajectory and is simultaneously the more *accurate* (median std-error vs. truth ≈ 0.003 vs. ≈ 0.02 for the short mode) and the more *efficient* (ESS/eval $\approx 4 \times 10^{-3}$ vs. 1×10^{-3}); the short mode is genuinely *stuck*, under-exploring a banana arm (Fig. 3b). The alarming part is that the stuck mode still passes the standard convergence check: its median rank-split $\hat{R}$ is only ≈ 1.01 and $\sim 87\%$ of stuck runs sit below the usual 1.03 threshold, so a practitioner watching $\hat{R}$ alone would accept a run with several-fold larger error and $\sim 4\times$ lower efficiency as converged.

Why is each mode *stable*? A state-dependent objective. The surprise is not that ChEES can pick a poor length once, but that the poor length is a *stable* attractor that longer tuning does not escape. The reason is that the ChEES objective is not fixed: it maximises ESJD in the *ensemble covariance* Σ^{-1} norm, and Σ is recomputed from the live walkers every step, so the length and the metric co-evolve. This closes a self-reinforcing loop with two fixed points. *Long*: a large L makes trajectories traverse and spread *along* the banana ridge, so Σ elongates along it, and ESJD in that elongated norm is maximised by long traversals — which keeps L large (IAT ≈ 1: one trajectory decorrelates). *Short*: a small L keeps the walkers local, so Σ stays compact and *under-dispersed* (hence the $\sim 8\times$ larger std-error), and in that compact norm even a short step already looks like a large normalised jump — which keeps L small (IAT ≈ 90). Tellingly, the two modes share essentially the *same* dual-averaged step size ($\epsilon \approx 0.03$) and acceptance (≈ 0.93): the step-size adapter and the acceptance criterion cannot tell them apart, only the length differs. Because ChEES ascends a *local* gradient (ADAM on $\log T$), it cannot tunnel between basins, and the co-adapting metric locks in whichever basin the early warmup fell into — classic multistability of an adaptive scheme whose objective depends on the state it has itself produced.

The sharp gap is a fingerprint of the Σ^{-1} norm. If the state-dependent metric is what sharpens the split, replacing it should blur the gap. It does. Re-running the d=2 study with ChEES measuring ESJD in a *fixed Euclidean* norm (`--chees-metric euclidean`, i.e. the identity matrix in place of Σ^{-1}; the h-walk move itself is unchanged) turns the cleanly bimodal $\bar{L}$ (short 3–30, long 123–160, empty in between) into a broad *continuum* ($\bar{L} = 4, 5, \ldots, 22, 38, 49, 62, 66, \ldots, 161$): the intermediate lengths that the Σ^{-1} feedback forbade are now populated. This is the predicted signature — the sharp two-attractor structure comes from the metric co-adapting with the length. But the Euclidean variant is not a fix: its efficiency and error are no better (it still produces short, stuck runs, because Euclidean ESJD is a poor proxy for mixing on the curved ridge), and it *forfeits the affine-invariance* that the Σ^{-1} norm exists to provide (§4a). The bistability is thus an intrinsic tension of trajectory-length adaptation on curvature, not an artefact of one norm choice.

It is not under-tuning. A natural hypothesis is that the short mode is simply a warmup that has not finished adapting. It is not (Fig. 3c): quadrupling the warmup ($8000 \to 32000$) leaves the long-mode fraction essentially unchanged, individual d=8 seeds stay stuck short at every warmup length tested (and some even flip long→short with *more* warmup), and varying the initialisation scale over $8\times$ does not move the fraction either. The basin a run lands in is a stochastic property of the coupled ensemble–ChEES dynamics on the curved geometry, not a transient. The practical remedy is therefore not "warm up longer" but **(i)** monitor an error/accuracy diagnostic (or the adapted $\bar{L}$ itself) alongside $\hat{R}$, which *does* separate the modes, and **(ii)** prefer a leaner ensemble, which we found lands in the long mode more often. That the whole phenomenon reproduces on the two-dimensional banana — where it costs seconds to study — makes it a clean, cheap test case for trajectory-length adapters on curvature.

4 Experiment 2 — conditioning vs. curvature

Setup. (a) Rotated Gaussian d=50, $\kappa = e^{2,4,6,8}$, warmup 1000, $8d$=400 walkers (the scaling rule); PEACHES (production 5000, tag `condf_lc`k), `emcee` (production 4×10^4), NUTS dense-mass (`condf_lc`k, the standard sampler) and NUTS identity-mass (`--nuts-mass identity`, production 2000, tag `condfid_lc`k, the ablation). (b) Rosenbrock $d \in \{8, 16, 32\}$, $8d$ walkers (tag `rz8`): PEACHES ($L_{\max} = 512$), `emcee`, NUTS identity-mass (tag `rz8id`). On Rosenbrock the global con-

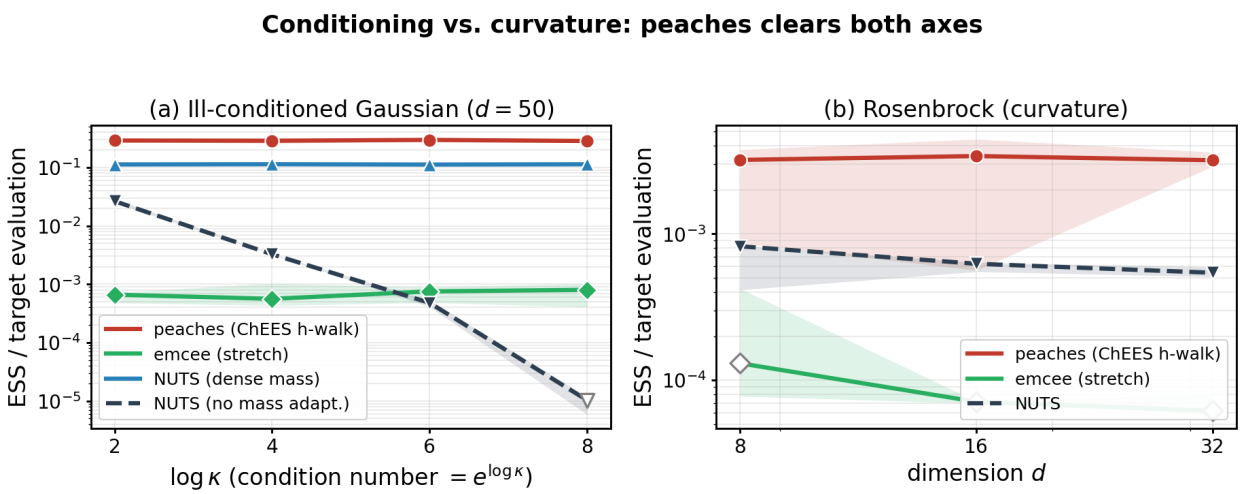


Figure 4: Two targets isolate two failure modes; only PEACHES clears both. Median over converged seeds, band = seed min–max; solid = converged ($\hat{R} \leq 1.03$), hollow = not. **(a) conditioning** — Gaussian d=50 vs. κ: PEACHES and `emcee` are flat because their metric is the ensemble covariance; NUTS is flat *only* through mass adaptation — fix its mass to the identity and efficiency collapses 4 orders. **(b) curvature** — Rosenbrock vs. d: the gradient methods (PEACHES, NUTS) integrate along the curved ridge and stay converged; `emcee`'s single affine map cannot, and it fails to converge for $d \geq 4$. Takeaway: the ensemble metric buys conditioning-robustness for free, gradient dynamics buy curvature-handling, and PEACHES has both.

ditioning is mild, so mass adaptation is irrelevant and the identity-mass NUTS represents NUTS faithfully. All runs are scored with the ensemble ESS.

Result — a clean 2×2. The two targets isolate two distinct failure modes. *Conditioning* (panel a): the empirical-covariance metric makes PEACHES and `emcee` affine-invariant, so their ESS/eval is flat in κ (≈ 0.29 and $\approx 7 \times 10^{-4}$, the latter just dimensionally penalised at d=50). NUTS is flat in κ *only* through its mass adaptation: with the mass fixed to the identity its ESS/eval falls from 2.6×10^{-2} ($\kappa = e^2$) to 5.9×10^{-6} ($\kappa = e^8$) — over four orders, and unconverged ($\hat{R} \approx 1.06$) at $\kappa = e^8$. *Curvature* (panel b): Rosenbrock's position-dependent curvature defeats any fixed linear map. `emcee`'s single affine stretch move cannot cross the ridge — even at 10^5 samples it fails to converge for $d \geq 4$ ($\hat{R}$ grows $1.14 \to 2.0$ over d=8 $\to$ 32). PEACHES and NUTS integrate *local* Hamiltonian dynamics along the ridge and stay converged (ESS/eval $\approx 10^{-3}$). Only PEACHES clears both axes: the ensemble preconditioner gives it the conditioning-robustness un-adapted NUTS lacks, and gradient dynamics give it the curvature-handling `emcee` lacks.

5 Experiment 3 — warmup robustness

Setup. Ill-conditioned Gaussian (d=100, $\kappa = e^6$, 128 walkers, tag `warmf_ww`) and Rosenbrock (d=8, 64 walkers, tag `warmfb_ww`); warmup $\in \{50, 100, 200, 500, 1000\}$, production 5000, 3 seeds, ensemble ESS. NUTS uses a dense mass matrix (the standard choice on an ill-conditioned target). PEACHES uses a 200-step stretch pre-warmup (§6) then its own warmup of the given length.

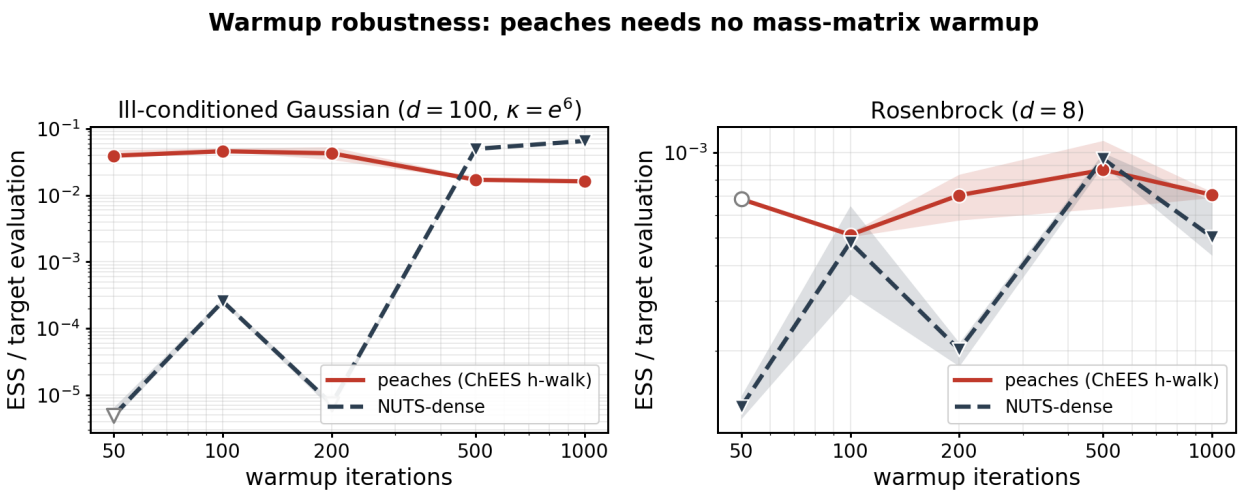


Figure 5: ESS/eval vs. gradient-sampler warmup length, 8 sampling seeds on a fixed target. Line = median over the converged seeds; band = interquartile range (25–75%) over those seeds (the IQR rather than min–max, because ChEES's integration length is bimodal on the banana). Walkers: 128 (Gaussian d=100), 64 (Rosenbrock d=8). PEACHES is flat from warmup 50 to 1000; NUTS-dense needs $\gtrsim 500$ warmup steps to fit its mass matrix and is orders less efficient below that (its d=8 Rosenbrock run at warmup 50 barely converges, shown hollow). Takeaway: because the metric is recomputed from the live ensemble every step, PEACHES has no mass matrix to warm up — the work NUTS spends in a long target-dependent warmup, it gets for free.

Result. Because the preconditioner is recomputed from the live ensemble every step, PEACHES needs no mass-matrix warmup: its ESS/eval is flat at ≈ 0.02–0.05 (Gaussian) across all warmup lengths. NUTS-dense must fit a $d(d{+}1)/2$-entry dense mass during warmup; at warmup 50 its ESS/eval is $\approx 5 \times 10^{-6}$ — a factor $\sim 10^4$ below PEACHES — and it becomes competitive only once warmup ≥ 500. The same pattern holds on Rosenbrock. This is the practical face of "the metric is free": the work NUTS spends in a long, target-dependent warmup, PEACHES gets for nothing.

6 Practical recommendation: a three-phase run

For PEACHES (and ensemble HMC generally) we recommend the following workflow, which the benchmarks above all use:

1. **Stretch pre-warmup (gradient-free).** Run a short `emcee`-style stretch move for w_s iterations *first*, from an over-dispersed initialization. This is cheap (one log-density per walker, no gradient) and affine-invariant, and it does two essential things: it moves the ensemble *mean* onto the typical set (a reasonable centre), and it shapes the ensemble *covariance* to the target's local geometry. This matters because PEACHES's preconditioner *is* the empirical covariance: an isotropic or collapsed initial cloud gives an ill-shaped metric, which in turn collapses the dual-averaged step size and wastes the gradient warmup. A short stretch phase suffices ($w_s \approx 100$–200; benefit saturates quickly); it uses no gradient and no knowledge of the target covariance.

2. **peaches warmup (gradient).** With the ensemble now well-centred and well-shaped, run the PEACHES adapters: `find_init_step_size` for the initial ϵ, dual averaging to the target

acceptance, and ChEES for the integration length. The empirical-covariance metric is re-estimated each step, so no mass matrix is fit here — only the scalar ϵ and the length L.

3. **Production.** Freeze (ϵ, L) and the adapters and collect samples. ESS/eval is measured here only.

In the code this is exactly `(warm_start` w_s`, warmup` w`, num_samples` N`)`. Phase 1 is the ensemble analogue of NUTS's mass-matrix warmup, but gradient-free and target-agnostic; phase 2 is much shorter and more robust than NUTS's because there is no mass matrix to converge (§5). Skipping phase 1 is the most common failure mode: without it the worst-mixing direction is initialization-dependent and the step size can collapse.

7 Reproducibility

All runs go through `python -m benchmark.run` (one `(target,dim,sampler,seed)` cell). Key flags: `--target` {`gaussian,rosenbrock`}, `--dim`, `--sampler` {`peaches,stretch,nuts`}, `--log-cond` ($\log \kappa$; $0 +$ `--no-rotate` for isotropic), `--n-chains`, `--warm-start` (stretch pre-warmup), `--warmup`, `--num-samples`, `--target-accept`, `--max-L` (PEACHES), `--nuts-mass` {`dense,diag,identity`}, `--max-tree-depth`, `--seed`, `--tag`.

Experiment	target	sampler configs	walkers / warmup
Scaling (§2)	$N(0, I_d)$, d=2–256	PEACHES (w_s=100); `emcee`	$8d$ (floor 64); 4k/5
Scaling (§2)	Rosenbrock d=2–32	PEACHES $L_{\max}$=512; `emcee`	$8d$ (floor 64); 8k/8
Bistability (§3)	Rosenbrock d=2 (& d=8)	PEACHES, warmup/init sweeps	64/256; up to 32k/
Conditioning (§4a)	Gauss d=50, κ=$e^{2\text{-}8}$	PEACHES; `emcee`; NUTS `dense/identity`	$8d$=400; 1k/5k
Curvature (§4b)	Rosenbrock $d \in \{8, 16, 32\}$	PEACHES $L_{\max}$=512; `emcee`; NUTS `identity`	$8d$ (floor 64); 8k/8
Warmup (§5)	Gauss d=100 κ=e^6; Rosen d=8	PEACHES (w_s=200); NUTS `dense`	400/64; {50..1000}

The scaling sweep uses 5 seeds (3 at d=256); the Rosenbrock scaling uses 8; the warmup experiment uses 6; the rest use 3. Per-dimension efficiency is the median over the converged seeds ($\hat{R} \leq 1.03$); the warmup figure's band is the interquartile range, all others are seed min–max. **All experiments are scored with the ensemble-mean ESS** (`--fast-ess`), so the whole note uses a single efficiency metric. Convergence ($\hat{R}$), bias, and std-error are stored alongside ESS/eval in each `metrics.json`. Figures are produced by `benchmark/plot_scaling2.py` (scaling) and `benchmark/plot_pn_*.py` and `plot_bistability.py` (the rest).

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